

RESEARCH INTERESTS

Programming languages, Compilers, Formal Verification, Security, Intermittent Computing, WebAssembly

EDUCATION

Carnegie Mellon University

Pittsburgh, USA

Ph.D. in Software Engineering, advised by Dr. Limin Jia

Aug 2023 – Present

Pohang University of Science and Technology (POSTECH)

Pohang, Korea

Bachelor of Computer Science and Engineering & Mathematics (double major)

Feb 2018 – Feb 2023

- GPA : 4.26/4.3 (Ranked 1st among all graduating students in class of 2023)
- Entered with presidential scholarship offered to one outstanding high school student in the year, and graduated with the highest GPA
- Early graduation by one-year (leave of absence for military duty during Jan 2020 – Dec 2021)

Korea Science Academy of KAIST (*High School for Science-Gifted Students*)

Busan, Korea

Graduated

Feb 2015 – Feb 2018

- GPA : 4.23/4.3 (Ranked 1st among all graduating students in class of 2018)

PROFESSIONAL SKILLS

Programming Languages: Rust, C++, SML, Python, WebAssembly, JavaScript, Scala, Maude, Futhark

Frameworks: LLVM, MLIR, React, Django, Git

ONGOING RESEARCH PROJECTS

Automatic Program Rescaling and Migration

April 2025 – Present

Programming language support for automatic program rescaling and migration

Hybrid CPU-GPU Task Parallelism

Dec 2023 – Present

Language support and task scheduling techniques for hybrid CPU-GPU task parallelism in MaPLe, SML-like functional language for multicore parallelism

- Developing multi-GPU support and task scheduling techniques to minimize overhead and maximize throughput

PUBLICATIONS

GraFeyn: Efficient Parallel Sparse Simulation of Quantum Circuits, Sam Westrick, Pengyu Liu, Byeongjee Kang, Colin McDonald, Mike Rainey, Mingkuan Xu, Jatin Arora, Yongshan Ding and Umut Acar, IEEE International Conference on Quantum Computing and Engineering (QCE) 2024 (**Best Paper**) (pdf)

Narrowing and Heuristic Search for Symbolic Reachability Analysis of Concurrent Object-Oriented Systems, Byeongjee Kang, Kyungmin Bae, Science of Computer Programming 2024 (doi.org/10.1016/j.scico.2024.103097)

Symbolic Reachability Analysis of Distributed Systems using Narrowing and Heuristic Search,

Byeongjee Kang, Kyungmin Bae, International Workshop on Formal Techniques for Safety-Critical Systems (FTSCS) 2022
(doi.acm.org?doi=3563822.3568017)

PREPRINTS

WAMI: Compilation to WebAssembly through MLIR without Losing Abstraction, Byeongjee Kang, Harsh Desai, Limin Jia, Brandon Lucia, Submitted to OOPSLA 2025

Fray: An Efficient General-Purpose Concurrency Testing Platform for the JVM, Ao Li, Byeongjee Kang, Vasudev Vikram, Isabella Laybourn, Samvid Dharanikota, Shrey Tiwari, Rohan Padhye, Submitted to OOPSLA 2025
(<https://arxiv.org/abs/2501.12618>)

Mr3: An Execution Framework for Hive and Spark Sungwoo Park, Seunggon Namgung, Byeongjee Kang, 2023
([dx.doi.org/10.2139/ssrn.4587845](https://doi.org/10.2139/ssrn.4587845))

TALKS

An MLIR Dialect for WebAssembly, WebAssembly Workshop 2025, [recording]

WORK EXPERIENCES

Microsoft Research

Redmond, WA, USA

Research intern at Data Systems Group advised by Dr. Kaushik Rajan

May 2025 – Aug 2025

- Worked on MLIR-based auto-vectorizing compiler with support for control flow dependencies

Kodebox

Seoul, Korea

Software engineer as military service alternative

Jan 2020 – Dec 2021

- **CodeChain Foundry**: Open-source blockchain engine based on composable module system
 - Worked as core developer of consensus engine
 - Implemented BLS signature aggregation scheme to optimize block seal size
 - Designed interface for modules to communicate with consensus engine
 - Tech Stacks: Rust, Docker, Kubernetes, Google Cloud Platform
- **Zuzu**: Web-based capturable management software
 - Worked as full-stack software engineer
 - Tech Stacks: Django (Python), React (TypeScript), GraphQL, Kubernetes, Google Cloud Platform

HONORS AND SCHOLARSHIPS

ILJU Fellowship (ILJU Academy and Culture Foundation)

Aug 2023 – July 2028

- Received total of USD 120,000 over eight semesters during graduate studies

Program for Highly Dedicated Students (POSTECH)

Feb 2018 – Feb 2023

- Presidential scholarship and supporting program in POSTECH offered to **one outstanding freshman** each year

Presidential Science Scholarship (Ministry of Education of Korea)

Feb 2018 – Feb 2023

- National science scholarship offered to top freshmen students in the field of science and technology

Global Leadership Program (CSE Department, POSTECH)

Sep 2019 – Present

- Scholarship and research support program for selected students in computer science department

Domun Scholarship (POSTECH)

Feb 2019

- Scholarship for freshmen with highest GPA

TEACHING EXPERIENCES

Student Mentoring Program (CSE Department, POSTECH)

Feb 2022 – June 2022

- Served as tutor for Algorithm course

Intensive Research Program (POSTECH)

Jan 2019

- Served as mentor in research experience program for science high school students

Last updated: May 7, 2025